

WHERE DID THE PROFIT GO?

Retail Chain Profit Leakage & Demand Optimization

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Executive Summary

This project investigated hidden profit leakage across sales, returns, inventory, vendors, and promotions. Although the retail chain generated █22.58M in net sales, multiple operational inefficiencies reduced profitability. Using SQL, Tableau, Power BI, and Excel, the analysis uncovered key leakage sources and developed actionable recommendations.

Metric	Value
Net Sales	█22.58M
Gross Profit	█3.58M
Gross Margin	15.86%
Return Rate	6.20%
Refund Amount	█1.64M
On-Time Delivery	69.63%

Key Insight: The business does not suffer from a revenue problem; it suffers from profit leakage across returns, inventory, vendor performance, and promotion effectiveness.

KEY FINDINGS

- ■1.64M was lost through refunds and return leakage.
- Returns were mainly driven by quality issues, damaged products, and late deliveries.
- 1,541 low-stock days and 1,292 stockout events created lost sales opportunities.
- Vendor delivery delays impacted inventory availability and customer satisfaction.
- Several promotions increased sales volume but reduced overall profitability.
- Profit leakage originated from multiple connected operational bottlenecks.

Root Cause Analysis

Vendor Delays → Stockouts → Customer Dissatisfaction → Returns → Refunds → Profit Leakage

Analytics Workflow

Data Collection → Data Cleaning → SQL Analysis → KPI Engineering → Tableau Visualization → Dashboard Development → Business Recommendations

RECOMMENDATIONS & BUSINESS IMPACT

- ✓ Improve vendor SLA performance and delivery reliability.
- ✓ Implement dynamic inventory replenishment and forecasting.
- ✓ Reduce return rates through stronger quality controls.
- ✓ Optimize promotional campaigns based on profitability.
- ✓ Adopt predictive analytics for future demand planning.

Expected Business Impact

Reduced refund leakage, improved inventory availability, higher customer satisfaction, stronger vendor performance, and increased profitability without requiring additional sales growth.

Conclusion

"The retail chain does not have a sales problem. It has a profit retention problem." This project transformed raw business data into actionable insights and identified opportunities to recover lost profits through operational improvements.